

Phase Mode Excitation of UCAs and Beamspace Notation

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- 1 Background
 - Signal Model
 - Steering Vector
 - ULA Manifold and Vandermonde Structure
- 2 UCA Manifold and Beamforming
- 3 Phase Mode Excitation and Beamspace Interpretation
- 4 Beamforming Matrices
 - Even Beamformer
 - Real Beamformer
- 5 References

Signal Model

- An incoming signal is modeled as a **uniform plane harmonic wave**:

$$s_n(t) = A \cos \left(2\pi f_c t + K_0 \hat{r} \cdot \vec{P}_n + \alpha \right)$$

where:

- $\vec{P}_n$ is the position vector of the n^{th} element
 - $K_0 = \frac{2\pi}{\lambda}$ is the wave number
 - $\hat{r}$ is the unit vector in the direction of arrival.
- In spherical coordinates:

$$\hat{r} = \begin{bmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{bmatrix} = \begin{bmatrix} u \\ v \\ \cos \theta \end{bmatrix} \quad (\text{direction cosines})$$

- Using complex notation:

$$s_n(t) = A e^{j\alpha} \cdot e^{jK_0 \hat{r} \cdot \vec{P}_n} \Rightarrow x_n(t) = \tilde{g}_0(t) \cdot e^{jK_0 \hat{r} \cdot \vec{P}_n}$$

where $\tilde{g}_0(t) = A e^{j\alpha}$ is the complex envelope at the origin.

Array Manifold or Steering Vector

- The **manifold** or **steering vector** is a length- N vector representing the response of an N -element array to a signal arriving from direction $\hat{r}$.
- It encodes the relative phase shifts across the array elements due to wave propagation:

$$\underline{\mathbf{a}}(\hat{r}) = \begin{bmatrix} e^{jK_0\hat{r}\cdot\vec{P}_0} \\ e^{jK_0\hat{r}\cdot\vec{P}_1} \\ \vdots \\ e^{jK_0\hat{r}\cdot\vec{P}_{N-1}} \end{bmatrix}$$

- **Array Pattern** is the sum of the responses described by the steering vector and is treated as the array output.

ULA Manifold and Vandermonde structure

- For an inter-element spacing of $d = \lambda/2$, and element position vector $\vec{P}_n = n \cdot d \cdot \hat{z}$, the array manifold looks like:

$$\underline{\mathbf{a}}(u) = \begin{bmatrix} 1 \\ e^{j\pi u} \\ e^{j2\pi u} \\ \vdots \\ e^{j(N-1)\pi u} \end{bmatrix} \quad \text{where } u = \cos(\theta)$$

- Now replace $e^{j\pi u}$ with a variable z :

$$\mathbf{a}(z) = [1, z, z^2, \dots, z^{N-1}]$$

- This is a **Vandermonde structure**, which allows the array response to be treated as a polynomial in z .
- Hence, the total array pattern becomes a polynomial — and by finding its roots, you can estimate the Direction of Arrival (DOA) [1].

UCA Manifold

- For a Uniform Circular Array (UCA) of radius r in the $\theta = 90^\circ$ plane (XY-plane), the position vector of the n^{th} element is:

$$\vec{P}_n = \begin{bmatrix} r \cos \gamma_n \\ r \sin \gamma_n \\ 0 \end{bmatrix}, \quad \text{where } \gamma_n = \frac{2\pi(n-1)}{N}$$

- Using complex envelope notation, the UCA steering vector becomes:

$$\underline{\mathbf{a}}(\theta, \phi) = \begin{bmatrix} e^{j\zeta \cos(\phi-\gamma_0)} \\ e^{j\zeta \cos(\phi-\gamma_1)} \\ \vdots \\ e^{j\zeta \cos(\phi-\gamma_{N-1})} \end{bmatrix}, \quad \text{where } \zeta = K_0 r \sin \theta$$

- The array data model is:

$$\mathbf{X}(t) = \mathbf{A} \cdot \mathbf{S}(t) + \mathbf{n}(t)$$

where $\mathbf{S}(t)$ is the signal vector at the reference point, and $\mathbf{n}(t)$ is the noise.

- **Beamforming** = weight and sum the sensor outputs to steer the array's response:

$$y(t) = \mathbf{w}^H \mathbf{X}(t)$$

Phase Mode Excitation

- Instead of weighting individual UCA elements arbitrarily, we define weights as a function of each element's angular position:

$$w(\gamma_n) = e^{jm\gamma_n}, \quad \text{where } \gamma_n = \frac{2\pi(n-1)}{N}$$

- m is the **mode** — an integer defining the spatial “harmonic”.

$$\text{Array output (mode } m) \Rightarrow \sum_{n=0}^{N-1} x_n(t) \cdot e^{jm\gamma_n}$$

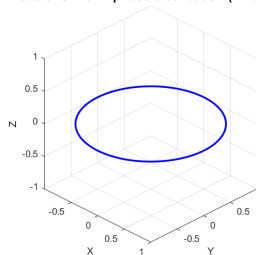
- Instead of synthesizing patterns element by element, you directly **excite the entire array into a mode** [2].
- **Analogy: Circular standing waves**
 - Mode 0: All weights equal — flat ring (DC component)
 - Mode 1: One full cycle — first spatial harmonic

⋮

- These are **spatial harmonics**, and just like in Fourier series, arbitrary patterns can be formed by superposition:

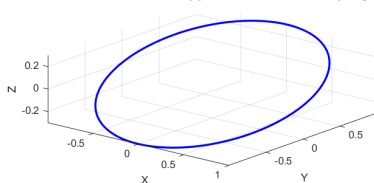
$$w(\gamma) = \sum_{m=-M}^M c_m e^{jm\gamma}$$

Mode 0: Uniform phase distribution (invariant)



Mode 0

Mode 1: Sine Wave Wrapped Around a Circle (1 Cycle)

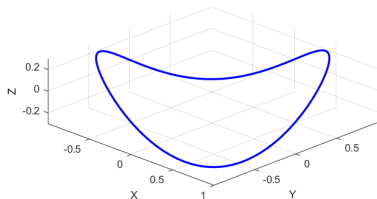


Mode 1

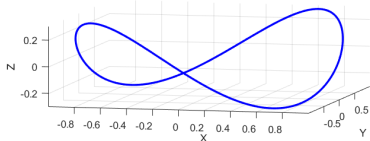
Each mode is a spatial harmonic. Higher modes increase the spatial frequency around the circle.

Higher Order Modes: 2 and 3

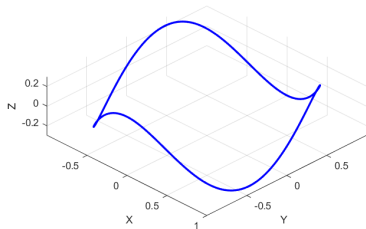
Mode 2: Sine Wave Wrapped Around a Circle (2 Cycles)



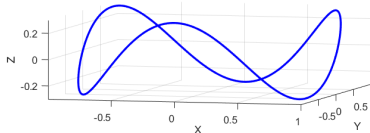
Mode 2: Sine Wave Wrapped Around a Circle (2 Cycles)



Mode 3: Sine Wave Wrapped Around a Circle (3 Cycles)



Mode 3: Sine Wave Wrapped Around a Circle (3 Cycles)



Beamspace Interpretation

- Applying mode- m weights to the UCA element outputs and summing gives the **beamspace array output**:

$$y_m(t) = \sum_{n=0}^{N-1} x_n(t) \cdot e^{jm\gamma_n}$$

- This operation transforms the array from **element space** (individual sensors) to **beamspace** (modal domain).
- You now treat the array as a whole — analyzing or steering **modes**, not elements.
- The far-field array pattern for mode m ($N > 2M$) becomes:

$$A_m(\phi) = j^m \cdot J_m(\zeta) \cdot e^{jm\phi}$$

where:

- $J_m(\zeta)$ is the Bessel function of the first kind (order m)
- $\zeta = k_0 r \sin \theta$, ϕ is the azimuth angle
- The exponential term $e^{jm\phi}$ introduces a **Vandermonde-like structure** in beamspace!

Beamforming Matrices

- The beamformer matrix $\mathbf{F}$ is designed such that:

$$\mathbf{F}\mathbf{F}^H = \mathbf{I} \quad \Rightarrow \quad (\text{orthogonal transformation preserves noise subspace})$$

- First, define the matrix $\mathbf{V}$ of modal weights:

$$\mathbf{V} = \sqrt{N} \begin{bmatrix} \mathbf{w}_{-M} & \cdots & \mathbf{w}_0 & \cdots & \mathbf{w}_M \end{bmatrix}$$

$$\text{where } \mathbf{w}_m = \frac{1}{N} \left[e^{jm\gamma_0}, \dots, e^{jm\gamma_{N-1}} \right]^T$$

- Its Hermitian:

$$\mathbf{V}^H \in \mathbb{C}^{(2M+1) \times N}$$

has each row as a beamformer for a spatial mode.

Even Beamformer $\mathbf{F}_e^H$

- Define the **even (complex) beamformer matrix**:

$$\mathbf{F}_e^H = \mathbf{C}_e \mathbf{V}^H, \quad \mathbf{C}_e = \text{diag}(j^{-M}, \dots, j^{-1}, 1, j^1, \dots, j^M)$$

- Beamspace manifold:

$$\mathbf{a}_e(\theta, \phi) = \mathbf{F}_e^H \mathbf{a}(\theta, \phi) = \sqrt{N} \cdot \mathbf{J}(\zeta) \cdot \mathbf{v}(\phi)$$

where:

- $\mathbf{J}(\zeta) = \text{diag}(J_M(\zeta), \dots, J_0(\zeta), \dots, J_M(\zeta))$
- $\mathbf{v}(\phi) = [e^{-jM\phi}, \dots, 1, \dots, e^{jM\phi}]^T$ is Vandermonde in $e^{j\phi}$
- This Vandermonde structure in beamspace enables **Root-MUSIC** and spectral search using FFT in **UCA-RB-MUSIC** [1].

Real-Valued Beamformer $\mathbf{F}_r^H$

- To obtain a real-valued beamspace representation, we define:

$$\mathbf{F}_r^H = \mathbf{W}^H \mathbf{F}_e^H \Rightarrow \mathbf{F}_r^H = \mathbf{W}^H \mathbf{C}_e \mathbf{V}^H$$

- The matrix $\mathbf{W}^H$ is composed of steering vectors in uniformly spaced directions:

$$\mathbf{W}^H = \frac{1}{\sqrt{M'}} \begin{bmatrix} \mathbf{v}(\alpha_{-M})^H \\ \vdots \\ \mathbf{v}(\alpha_0)^H \\ \vdots \\ \mathbf{v}(\alpha_M)^H \end{bmatrix}, \quad \alpha_i = \frac{2\pi i}{M'} \quad (i = -M, \dots, M)$$

- The real-valued beamspace manifold is:

$$\mathbf{a}_r(\theta, \phi) = \mathbf{F}_r^H \mathbf{a}(\theta, \phi) = \sqrt{N} \cdot \mathbf{W}^H \mathbf{J}(\zeta) \mathbf{v}(\phi)$$

- The output can be expressed as:

$$\mathbf{a}_r(\theta, \phi) = \begin{bmatrix} f(\zeta, \phi - \alpha_{-M}) \\ \vdots \\ f(\zeta, \phi - \alpha_0) \\ \vdots \\ f(\zeta, \phi - \alpha_M) \end{bmatrix} \quad (\text{real-valued spatial beams})$$

- Where:

$$f(\zeta, \phi) = \sqrt{\frac{N}{M'}} \left[J_0(\zeta) + 2 \sum_{m=1}^M J_m(\zeta) \cos(m\phi) \right] \in \mathbb{R}$$

References I

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-  J. Davis and A. Gibson, “Phase mode excitation in beamforming arrays,” in *Proceedings of the 3rd European Radar Conference (EuRAD)*. Manchester, UK: EuMA, 2006, pp. 307–310.